A

Mini Project Report

on

SideKik - A freelancing platform using NLP and ANN

Submitted in partial fulfillment of the requirements for the degree

Third Year Engineering – Computer Science Engineering (Data Science)

by

Siddesh Patil 22107031

Raj Choudhary 22107044

Varun Lad 22107043

Devesh Patil 22107038

Under The Guidance Of

Mr. Praveen Shinde

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING (DATA SCIENCE)

A.P. SHAH INSTITUTE OF TECHNOLOGY

G.B. Road, Kasarvadavali, Thane (W)-400615

UNIVERSITY OF MUMBAI

Academic year: 2024-25

CERTIFICATE

This to certify that the Mini Project report on SideKik - A Freelancing Platform using NLP & ANN has been submitted by Siddesh Patil (22107031), Raj Choudhary (22107044), Varun Lad (22107043), Devesh Patil (22107038), who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2024-2025** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

Mr. Praveen Shinde
Guide

Ms. Anagha Aher
HOD, CSE(Data Science)

Dr. Uttam D. Kolekar
Principal

External Examiner:
1.

Internal Examiner:
1.

Place: A. P. Shah Institute of Technology, Thane
Date:

ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide **Mr. Praveen Shinde**.Expressing gratitude towards our HoD, **Ms. Anagha Aher**, and the Department of Computer Science Engineering (Data Science) for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our project coordinator **Ms. Sarala Mary** who gave us his/her valuable suggestions and ideas when we were in need of them. We would also like to thank our peers for their helpful suggestions.

TABLE OF CONTENTS

ABSTRACT

Freelancing platforms often struggle with inefficient talent matching, lack of personalization, and limited earning opportunities for freelancers. Side-Kik addresses these challenges by leveraging Natural Language Processing (NLP) and Approximate Nearest Neighbors (ANN) algorithms to provide AI-driven freelancer recommendations based on project descriptions. Unlike traditional platforms that rely on basic keyword searches, Side-Kik enhances the client experience with personalized search options using FAISS and ElasticSearch. Additionally, the platform introduces a Pre-Built Project Marketplace, enabling freelancers to sell completed projects and create new revenue streams. To help users stay informed about industry trends, Side-Kik integrates real-time market analysis using the Adzuna API. By combining AI-powered recommendations, dynamic search capabilities, and an innovative freelancer marketplace, Side-Kik streamlines the hiring process and revolutionizes the freelancing ecosystem..

Chapter 1

Introduction

Freelancing has become an essential part of the global economy, enabling businesses to access skilled professionals while allowing individuals to work flexibly. However, existing freelancing platforms often struggle with inefficient talent matching, lack of personalization, and limited earning opportunities for freelancers beyond direct client projects.

To address these challenges, Side-Kik introduces an AI-powered freelancing platform designed to enhance the freelancer-client matching process. By leveraging Natural Language Processing (NLP) and Approximate Nearest Neighbors (ANN) algorithms, the platform analyzes project descriptions and recommends the most suitable freelancers.

Additionally, Side-Kik features a Pre-Built Project Marketplace, allowing freelancers to sell completed projects, creating a new revenue stream. With FAISS and ElasticSearch, users benefit from a highly personalized search experience, while real-time market trend analysis powered by the Adzuna API ensures they stay updated on in-demand skills.

1.1. Purpose:

The purpose of Side-Kik is to transform the freelancing landscape by addressing the inefficiencies of traditional hiring platforms and providing a seamless, AI-driven experience for both clients and freelancers.

Intelligent Freelancer Matching: By leveraging Natural Language Processing (NLP) and Approximate Nearest Neighbors (ANN), Side-Kik ensures precise freelancer recommendations based on project descriptions, reducing the time and effort required for clients to find the right talent.

Personalized Search Experience: Unlike conventional keyword-based searches, the platform integrates FAISS and Elastic Search to deliver highly personalized and relevant freelancer suggestions, enhancing user satisfaction.

Pre-Built Project Marketplace: To empower freelancers with additional income opportunities, Side-Kik enables them to sell completed projects, offering clients immediate access to high-quality solutions.

Real-Time Market Insights: By integrating the Adzuna API, the platform provides users with real-time market trends, helping freelancers stay updated on in-demand skills and guiding clients toward informed hiring decisions.

Seamless and Engaging User Experience: With an intuitive interface, AI-driven recommendations, and marketplace functionalities, Side-Kik streamlines freelancer-client interactions, fostering a dynamic and efficient digital freelancing ecosystem.

1.2. Problem Statement:

Finding the right freelancer for a project can be time-consuming and inefficient, as existing platforms rely on manual searches and lack intelligent recommendations. Additionally, freelancers have limited avenues to monetize their pre-built projects, and users struggle to access real-time market trends to make informed decisions.

Solution:

Our platform leverages AI-driven recommendations to match clients with the most suitable freelancers based on project descriptions. It introduces a Pre-Built Project Marketplace where freelancers can sell completed projects, providing an additional revenue stream. The platform also offers a Personalized Search System for efficient talent discovery and integrates Market Trends Analysis using APIs to help users stay updated on in-demand skills.

1.3. Objectives:

To develop an AI-Driven Freelancing Platform: Side-Kik aims to streamline freelancer-client matching using NLP and ANN (FAISS) for intelligent recommendations. It enhances search efficiency with FAISS and Elastic Search, enabling personalized freelancer discovery. The platform also introduces a Pre-Built Project Marketplace, allowing freelancers to sell completed projects, and integrates Adzuna API for real-time market trend analysis, helping users stay updated on in-demand skills.

Develop an AI-Driven Freelancer Matching System: Utilize NLP and ANN (FAISS library) to recommend freelancers based on project descriptions, improving accuracy and efficiency in talent selection.

Implement Personalized Search and Filtering: Leverage FAISS and Elastic Search to provide clients with tailored freelancer recommendations based on their project requirements.

Enable a Pre-Built Project Marketplace: Allow freelancers to sell completed projects, offering an additional revenue stream and providing clients with ready-made solutions.

Integrate Real-Time Market Trends Analysis: Use the Adzuna API to display in-demand skills and job trends, helping users make informed decisions.

1.4. Scope:

The scope of the Side-Kik platform outlines the key functionalities and innovations aimed at streamlining freelancer-client interactions, improving service discovery, and ensuring secure, scalable platform management. The major focus areas include:

AI-Powered Freelancer Recommendations: Implement intelligent talent matching using NLP and ANN algorithms to connect clients with the most suitable freelancers.

Advanced Search Capabilities: Enhance freelancer discovery through personalized search filters powered by FAISS and Elastic Search.

Pre-Built Project Selling Feature: Develop a marketplace where freelancers can upload and sell projects, providing clients with instant access to solutions.

Secure Authentication and Data Management: Utilize Supabase for user authentication, database management, and project file storage, ensuring data security and scalability.

Chapter 2

Literature Review

The rise of freelancing platforms has revolutionized the global job market by providing remote work opportunities, flexible employment, and seamless client-freelancer interactions. Various research studies have examined the effectiveness, challenges, and technological advancements in freelancing platforms. This literature review explores recent developments, focusing on platform architecture, user experience, security concerns, and AI-based recommendations.

Platform Architecture and Functionalities: Several freelancing platforms have been designed with a focus on enhancing user experience, efficient project matching, and secure payment integration. Aditya Pawar et al. [1] developed "FrJobs," a freelancing website that streamlines job posting, proposal submission, and secure communication between clients and freelancers. Their study highlights the importance of a well-structured platform in ensuring smooth transactions and engagement. Similarly, Reshma Totare et al. [2] introduced "Takenoko," which incorporates project categorization and user rating mechanisms, ensuring a fair and transparent freelancing ecosystem.

User Experience and Platform Engagement: User satisfaction is a critical factor in freelancing platforms. Amit Narote et al. [3] developed a web application tailored for freelancing developers and designers, emphasizing intuitive UI/UX design and efficient search functionalities. Their study revealed that freelancers prefer platforms with easy navigation, project filtering, and interactive dashboards. Additionally, Bhupinder Kaur et al. [4] analyzed the impact of customer feedback systems on freelancer credibility and client trust, underscoring the necessity of real-time review mechanisms to enhance engagement.

Security Concerns in Freelancing Websites: Ensuring secure transactions and data protection is a primary concern in freelancing platforms. Studies emphasize the importance of encrypted communication and fraud detection mechanisms. Mr. Akshaya K M et al. [5] proposed a freelancing platform integrating advanced authentication and escrow-based payment systems to mitigate scams and ensure fair compensation. Their research highlights the increasing need for blockchain-based solutions to enhance transactional transparency and prevent fraudulent activities.

AI-Based Matching and Recommendation Systems: AI-driven recommendation systems are transforming freelancing platforms by improving job-freelancer matching accuracy. Narote et al. [3] discussed the integration of AI in skill-based job allocation, ensuring optimized task distribution. Moreover, Pawar et al. [1] explored machine learning techniques for predicting project success rates based on freelancer experience and client requirements. These studies indicate that incorporating AI enhances hiring efficiency and provides freelancers with personalized job recommendations.

Challenges and Future Directions: Despite significant advancements, freelancing platforms face challenges such as payment disputes, skill misrepresentation, and low project retention rates. Kaur et al. [4] pointed out that freelancer credibility remains a major issue, often leading to mismatches and dissatisfaction. Totare et al. [2] suggested that integrating blockchain for identity verification and smart contracts could improve platform reliability. Future research should focus on developing automated dispute resolution mechanisms and incorporating AI-driven skill assessments for better freelancer evaluation.

Chapter 3

Proposed System

Side-Kik is an AI-powered freelancing platform designed to enhance freelancer-client connections through intelligent recommendations and seamless project transactions.

Key features of Proposed System include:

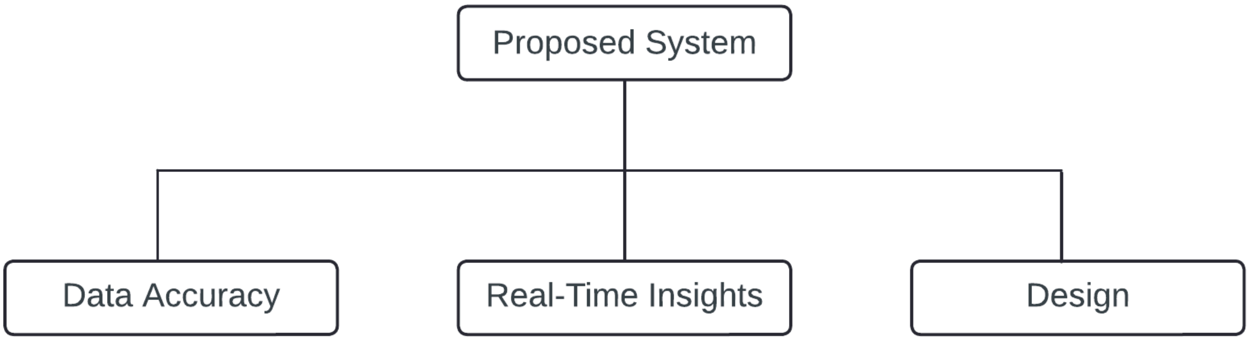

Figure 3.1: Proposed System

Data Accuracy: Ensures precise freelancer matching by analyzing project descriptions using advanced AI models, improving the quality of recommendations.

Real-Time Market Insights: Utilizes the Adzuna API to display trending skills and job market trends, helping users stay competitive.

User-Friendly Interface and Experience: The MoodTracker system will be designed with a clean, intuitive user interface to ensure accessibility for all users.

3.1. Features/Functionality:

Side-Kik integrates AI-driven technologies to enhance the freelancing experience by providing intelligent recommendations, personalized searches, and seamless project transactions. The platform ensures efficiency, security, and accessibility for both freelancers and clients.

AI-Powered Freelancer Matching: Utilizes NLP and FAISS-based ANN algorithms to analyze project descriptions and suggest the most relevant freelancers. This ensures accurate and efficient freelancer-client matching, improving hiring success rates.

Personalized Search and Filtering: Incorporates FAISS and Elastic Search to provide customized search options based on user preferences, freelancer expertise, and project requirements, making talent discovery faster and more precise.

Pre-Built Project Marketplace: Allows freelancers to sell completed projects, enabling clients to purchase ready-made solutions. This streamlines project acquisition and provides an additional revenue stream for freelancers.

Market Trends Analysis: Leverages the Adzuna API to display real-time skill demand trends and job market insights, helping freelancers stay updated on industry needs and adjust their skills accordingly.

User-Friendly Dashboard:A well-structured dashboard for both freelancers and clients, offering an intuitive interface for managing projects, applications, earnings, and transactions efficiently.

Chapter 4

Requirement Analysis

Requirement analysis is a crucial phase in developing Side-Kik, ensuring the platform aligns with freelancer and client needs. This phase defines the system's scope, core functionalities, database structure, and technology stack to build an efficient and user-friendly freelancing marketplace.Side-Kik caters to freelancers and clients, enabling talent discovery, project collaboration, and secure transactions. Freelancers use the platform to showcase skills, upload pre-built projects, and receive personalized client recommendations, while clients leverage AI-driven suggestions to find the best freelancers and purchase pre-built solutions.

Functional requirements include an AI-driven freelancer matching system that accurately connects clients with freelancers based on their skills and project descriptions. A prebuilt project marketplace allows freelancers to upload completed projects, enabling clients to purchase ready-made solutions securely. Market trends analysis provides real-time insights into high-demand skills, helping freelancers stay competitive. Additionally, a secure authentication and profile management system powered by Supabase ensures smooth login/signup, while freelancers can manage their skills, experience, and availability within their profiles.

Non-functional requirements such as scalability, security, and user experience are crucial for Side-Kik's success. The platform must efficiently handle high user traffic and concurrent transactions to ensure seamless interactions. Security is reinforced through Supabase authentication, protecting user data. A clean, intuitive user interface enhances usability, ensuring smooth navigation for both freelancers and clients.

The database schema is a core component of Side-Kik, defining key entities such as Users, Freelancer Profiles, Projects. The Users table differentiates between clients and freelancers, managing authentication and profile details. Freelancer Profiles store relevant details like skills, experience, and availability for accurate recommendations.

Chapter 5

Project Design

Project design refers to the process of conceptualizing and planning the structure, components, and functionalities of a project to achieve specific objectives. It involves translating the requirements and goals identified during the initial phases (such as requirement analysis) into a detailed blueprint or roadmap for implementation.

5.1 Use Case Diagram:

It is a visual representation that models the interactions between users (or other systems) and a system, describing its functionality and behavior from the user's perspective.

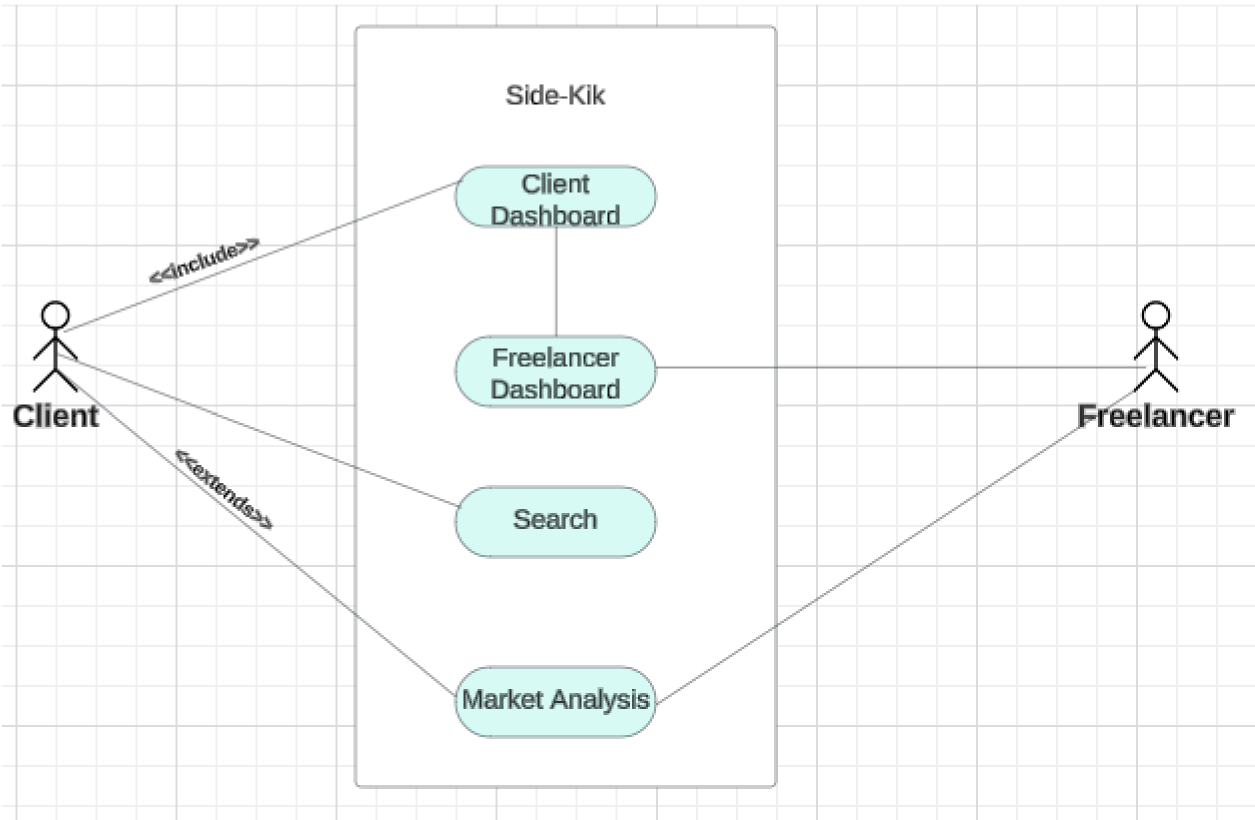

Figure 5.1: Use Case Diagram

In Fig 5.1 , Central Character is User and the diagram shows user interactions with Application. Diagram explains 2 components.

1. Actors (Clients and Freelancers): The users are the primary actors who interact with the system.

2.Interaction: The User can interact with the provided features.

5.2 DFD (Data Flow Diagram):

It is a graphical representation of the flow of data within a system, showing how data moves between processes, external entities, and data stores. It's widely used in system analysis and design to model the logical flow of information through a system.

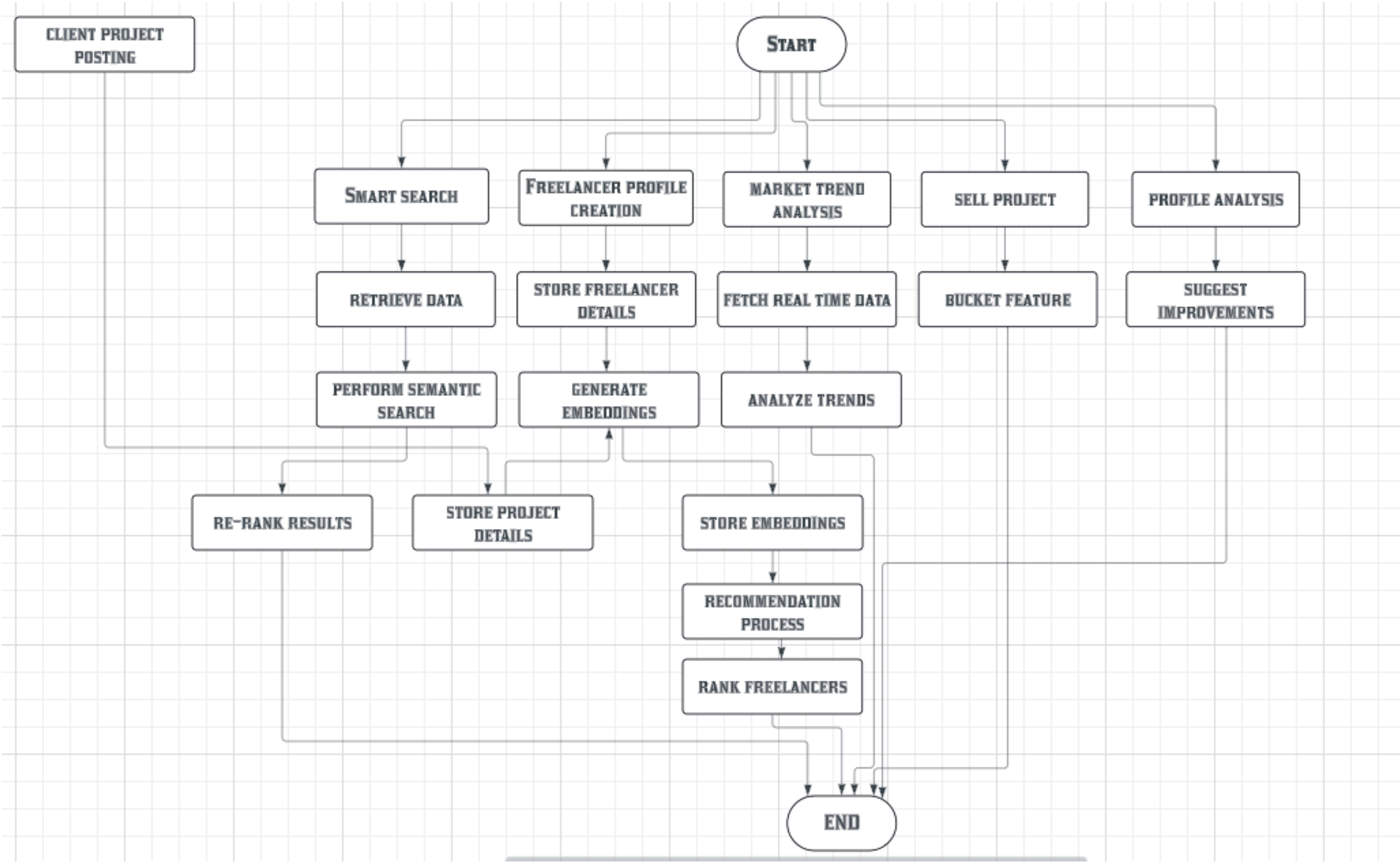

Figure 5.2: Data Flow Diagram

The project architecture depicted in the Data Flow Diagram (DFD) outlines a seamless workflow for Side-Kik.This flowchart outlines the end-to-end workflow of a freelance platform, beginning with inputs like client project postings, freelancer profiles, market trend analysis, and project uploads. Smart search and semantic search use retrieved data and embeddings to match projects with suitable freelancers. Freelancer details and market trends are processed to generate embeddings, which are stored and used in the recommendation and ranking process. The system also includes features for selling pre-built projects and analyzing profiles for improvement. All modules integrate to provide ranked freelancer recommendations, project suggestions, and insights—ensuring a seamless and intelligent user experience.

5.3 System Architecture:

It refers to the conceptual design that defines the structure and behavior of a system. It provides a blueprint for how the system's components and subcomponents interact with each other and with external systems to achieve the desired functionality.

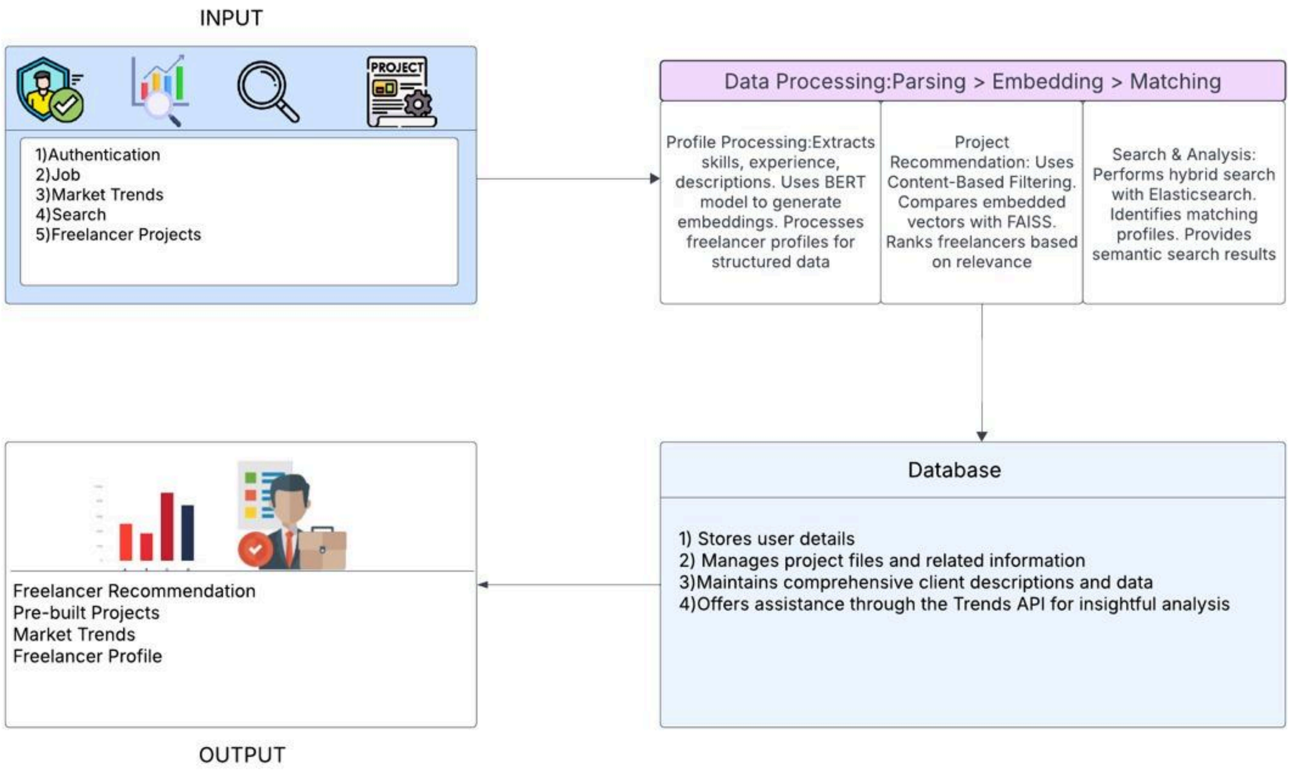

Figure 5.3: System Architecture

This system is designed to intelligently recommend freelancers, analyze market trends, and match project requirements using advanced data processing and machine learning techniques. It begins with multiple input types including user authentication, job postings, market trend data, search queries, and freelancer project details, which form the foundation for analysis and recommendation. At its core lies a robust data processing pipeline involving parsing, embedding, and matching. Freelancer profiles are parsed to extract structured data such as skills, experience, and descriptions, which are then embedded using a BERT model to capture semantic meaning. These embeddings are used in a content-based filtering approach, where a FAISS-powered engine compares project descriptions with freelancer embeddings to rank and recommend the most suitable candidates. Additionally, the system supports semantic search via Elasticsearch, enabling context-aware and relevant freelancer and project discovery. All processed data is stored in a centralized database that holds user profiles, project files, and client data, while also integrating with external APIs like Google Trends to extract insights into emerging market demands. The system ultimately delivers intelligent output including freelancer recommendations, pre-built project suggestions, current market trends.

5.4 Implementation:

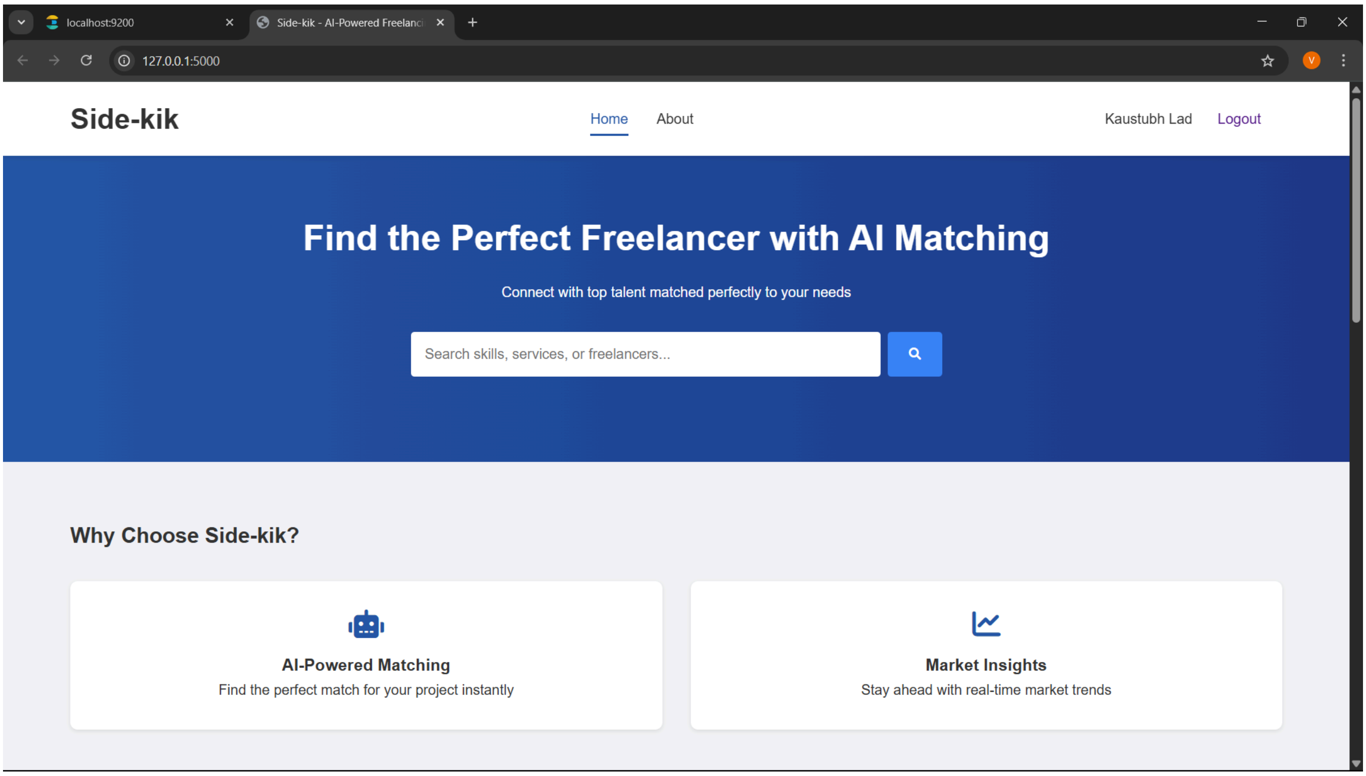

Figure 5.4.1: Client-Dashboard

The above figure shows the client-dashboard side when the user logs in as a client.It displays all available features tailored for client interaction and navigation.

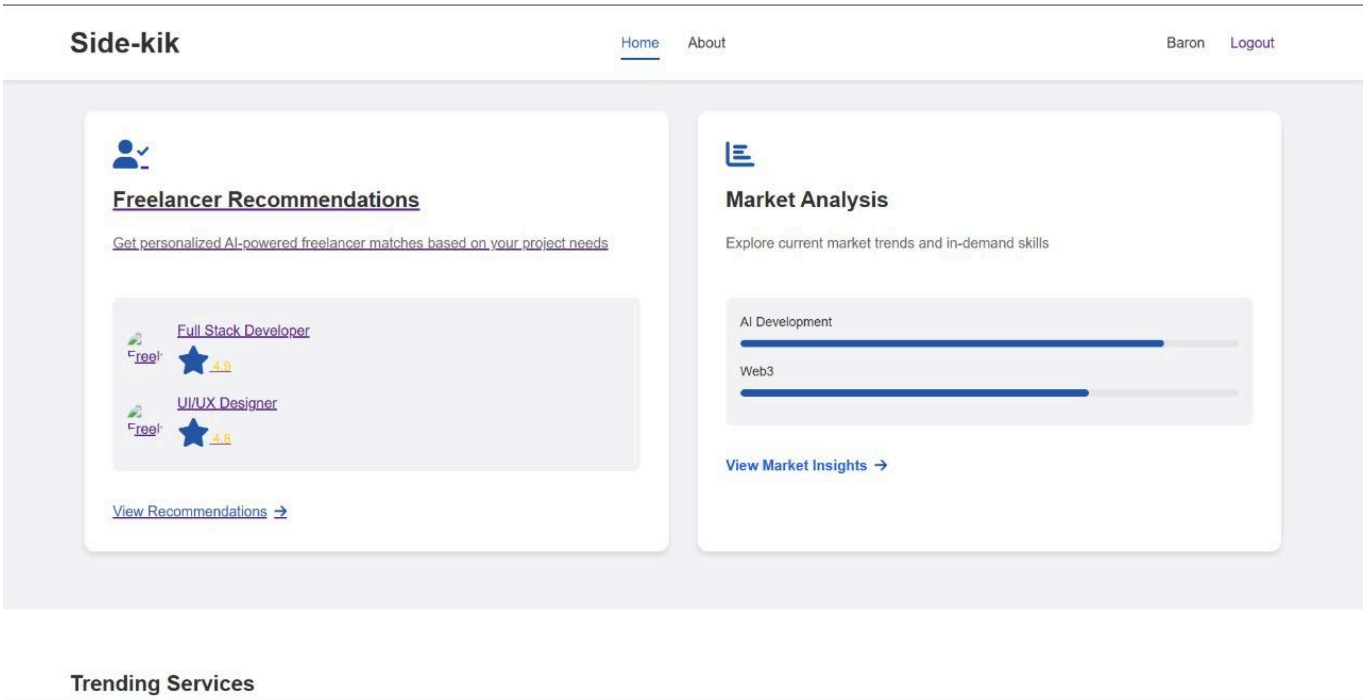

Figure 5.4.2: Client-Dashboard-Feature

Above figure shows the features of the client-dashboard.It highlights options like project posting,recommendations, and trend analysis.

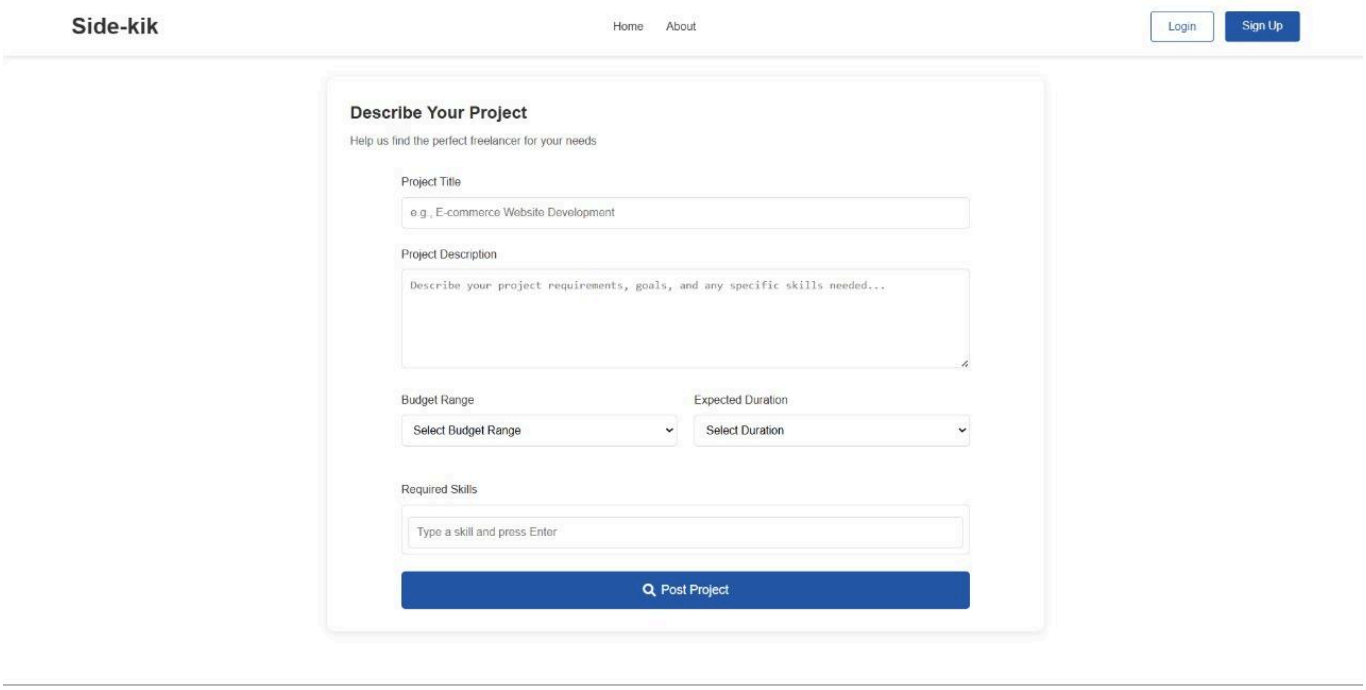

Figure 5.4.3: Freelancer Recommendations(A)

The above figure shows the freelancer-recommendations page where clients can search for freelancers according to their job needs.Clients can input job details to get a list of matching freelancers.

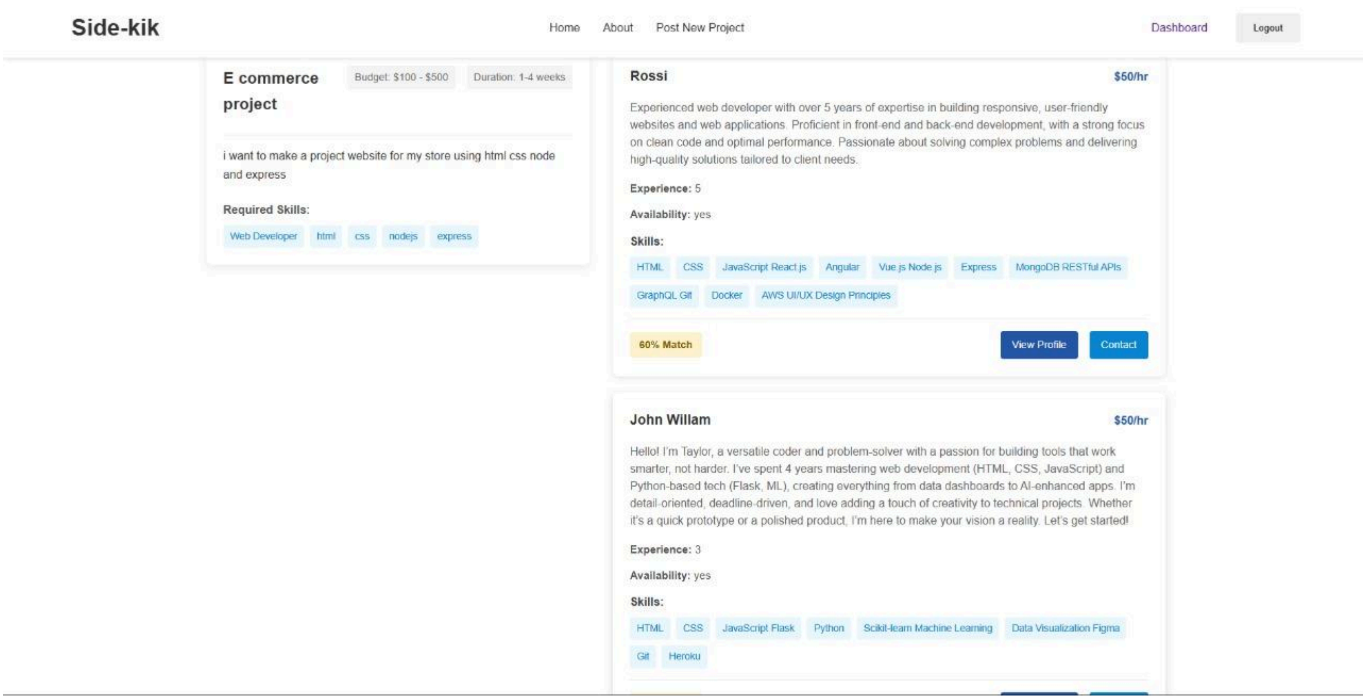

Figure 5.4.4: Freelancer Recommendations(B)

The above figure shows the recommended freelancers according to the job posting of the client.It displays a ranked list of freelancers best suited for the client's requirements.

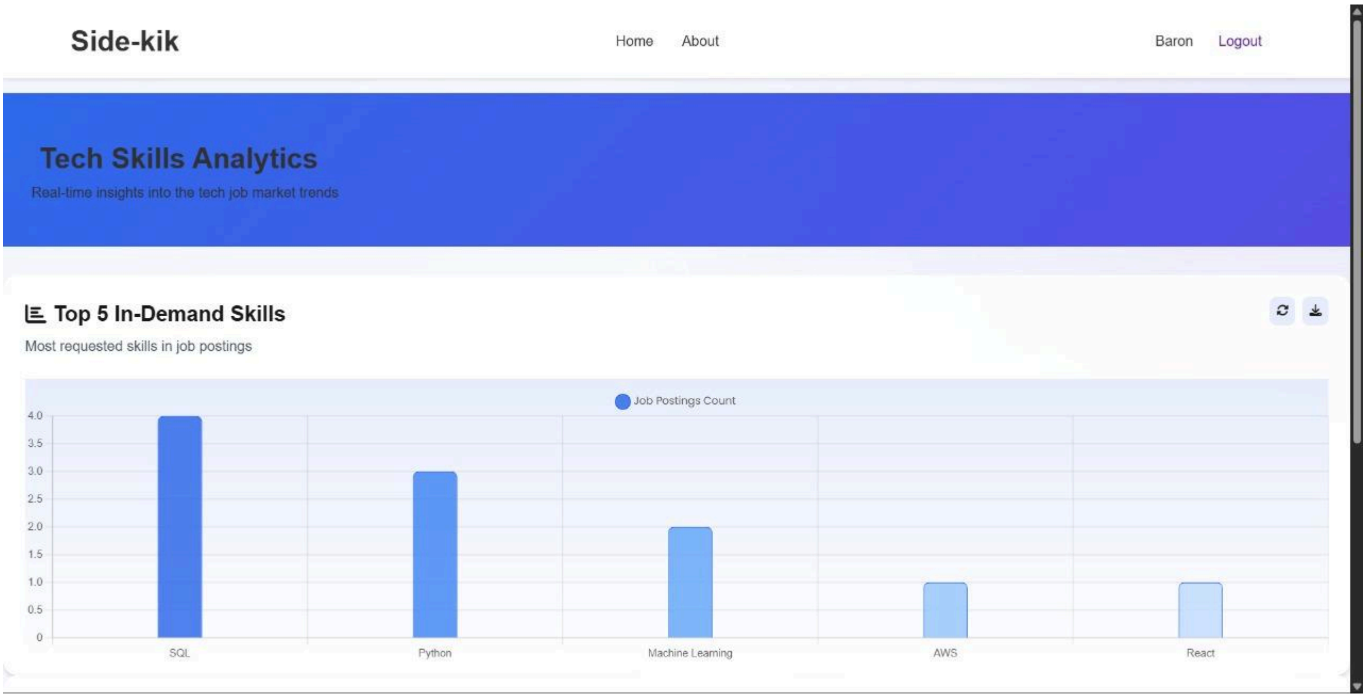

Figure 5.4.5: Market Analysis

The above figure shows the current market trends to the user.It provides insights into trending skills and in-demand technologies.

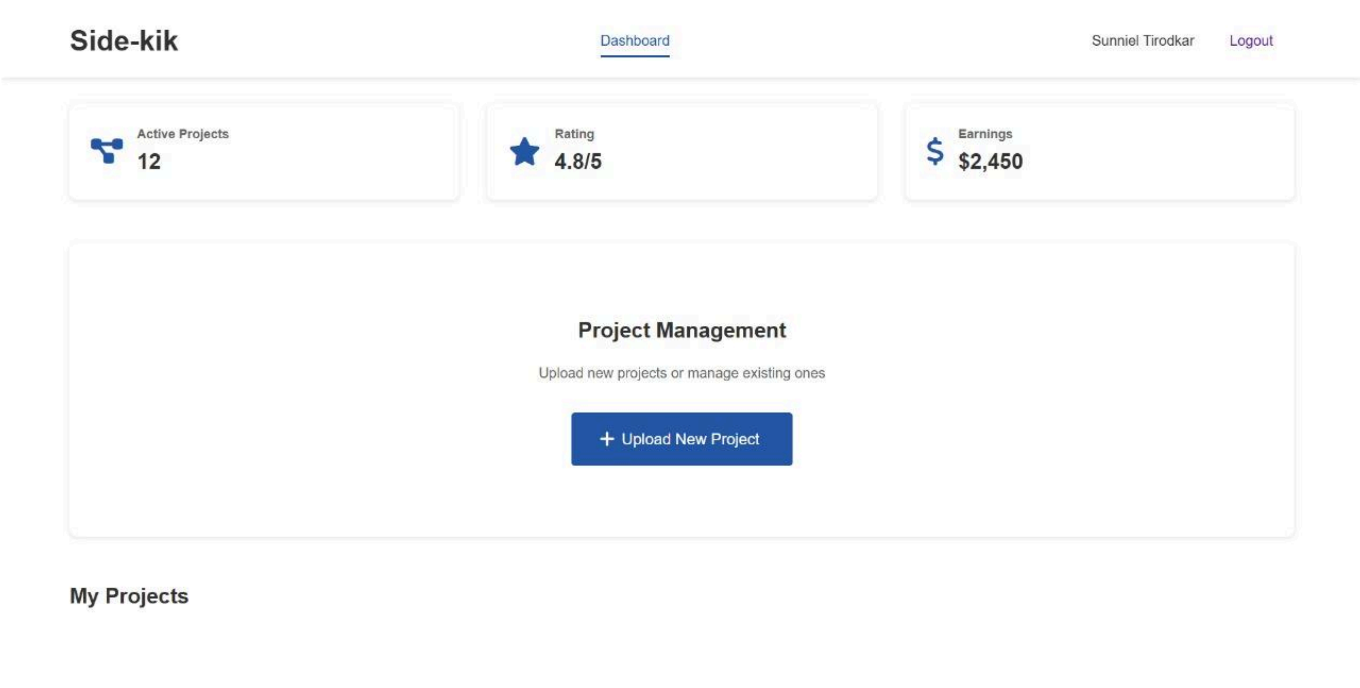

Figure 5.4.6: Freelancer-Dashboard

The above figure shows the freelancer-dashboard side when the user logs in as a freelancer.It includes features like profile updates, project uploads.

Chapter 6

TECHNICAL SPECIFICATION

The technical specifications provide a detailed outline of the necessary tools, technologies, and infrastructure needed to execute the project effectively. In our project, these specifications encompass the selection of programming languages to ensure that the project is equipped with the appropriate resources for compatibility, scalability, and efficiency throughout its development and deployment phases.

Front-End (User Interface):

1.HTML5 + CSS3: Used for structuring and styling the web pages. HTML5 ensures semantic markup, while CSS3 is used to design a clean, responsive interface.

2.JavaScript ES2024: Implements interactivity in the front-end, handling real-time updates, form validation, and dynamic rendering of freelancing projects and profiles.

Back-End (Server & Data Handling):

1. **Python 3.12.1**: Serves as the primary language for back-end development, handling data processing, AI-based recommendations, and API integration.
2. **Flask Framework 8.0**: Acts as the back-end framework, managing HTTP requests, routing, and server-side logic. Flask also connects the front-end with machine learning models and APIs.
3. **Elasticsearch 8.15.0**: Added for full-text search, complementing FAISS's vector search. Enables keyword-based personalized search
4. **Adzuna API**: Integrated to fetch real-time job market data and trends for freelancers.

Database:

1.Supabase (PostgreSQL-based DB): Stores freelancer profiles, project listings, client requests, and historical interactions to enhance AI-driven recommendations.

Machine Learning (AI-Based Recommendations & Search Optimization):

1. **Datasets:**
   - **Synthetic Freelancers Dataset**: 10,000 rows
   - **Synthetic Projects Dataset**: 7,000 rows These datasets are used to train AI models for freelancer-project matching and recommendations.
2. **Algorithms:**
   - **Natural Language Processing (NLP) - BERT Embeddings**: Utilized for semantic analysis and skill-based project matching using sentence-transformers/all-MiniLM-L6-v2.
   - **Approximate Nearest Neighbors (ANN) - FAISS**: Used for fast and scalable

freelancer-project matching based on skill embeddings.

3. **ML Libraries:**
   - **Hugging Face**: For implementing pre-trained BERT models for text embeddings.
   - **FAISS**: For efficient similarity search and recommendation systems.
   - **NumPy & Pandas**: For numerical operations and data manipulation during preprocessing and analysis.
   - **Scikit-learn 1.5.2**: For evaluation metrics (precision, recall, F1) and dataset splitting.
4. **Data Preprocessing:**
   - **Text Cleaning**: Removing punctuation, URLs, numbers, and stopwords for optimal NLP model performance.
   - **Tokenizing & Padding**: Ensures uniform input structure for AI models.
   - **Dataset Splitting**: Data is split into training and testing sets for model validation.

Development Environment & Version Control:

1. **VS Code 1.93.1**: Used as the integrated development environment (IDE) for writing, testing, and debugging code.
2. **Git & GitHub**: For version control, collaboration, and maintaining code repositories.
3. **Jupyter Notebook 7.2.1**: Used for data exploration, training ML models, and visualization

Chapter 7

Project Schedule

In our project, the Gantt chart will outline key activities where each task will be represented by a bar on the chart, indicating its start and end dates, duration, and dependencies, allowing project stakeholders to track progress, identify potential delays, and timely completion of project objectives.

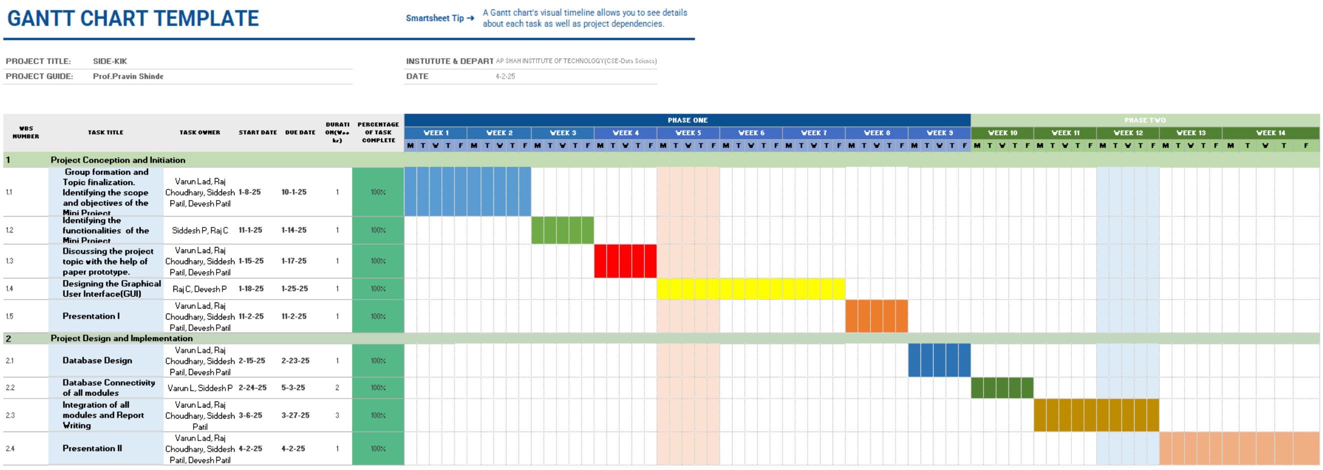

Figure 7.1: Gantt Chart

The project conception and initiation task were executed by the start of the month around 08/01/25. The task of initiation included many more sub-tasks such as group formation and topic finalization which was performed during the 1st week of project initialization.

The group formed included 4 members Varun Lad, Raj Choudhary, Siddesh Patil, Devesh Patil and the finalized topic was Side-Kik.Further, the upcoming week led to the task of identifying the scope and objectives of the mini projects. The next sub-task was to identify the functionalities of the project which was done by all the members in a span of 11/01/25 to 14/01/25. The discussion of the project topic with the help of a paper prototype was completed with equal contribution from all the group members within 15/01/25-17/01/25.

The next task, Database Connectivity and functionalities of modules in the app were done by Varun Lad, Raj Choudhary, Siddesh Patil, Devesh Patil from 24/02/25 to 05/03/25. The Integration of all modules, user interfaces and report writing was completed by Varun Lad, Raj Choudhary, Siddesh Patil, Devesh Patil from 06/03/25 to 27/03/25. The preparation of final presentation II work was equally shared by all the group members in the time of 02/04/2025.

Chapter 8

Results

The project results section provides an overview of the outcomes achieved through the development and implementation of Side-Kik. This section highlights the system's key functionalities, technical aspects, and overall impact, offering valuable insights into its effectiveness in connecting freelancers with clients and facilitating seamless transactions.

System Overview:

Side-Kik is a comprehensive freelancing platform built using Flask and integrated with Supabase for authentication, database management, and secure file storage. The platform enables clients to search for freelancers based on skills, view freelancer profiles, and purchase pre built projects securely. An AI-powered freelancer recommendation system enhances project-to-freelancer matching, improving the efficiency of hiring. Freelancers can create detailed profiles and upload pre-built projects. The system also provides real-time market trends analysis, allowing freelancers to stay updated on in-demand skills. With an intuitive user interface, Side-Kik ensures a seamless experience for both clients and freelancers, streamlining the freelancing process.

System Architecture:

Side-Kik follows a modular architecture, with Flask serving as the backend framework to manage authentication, profile management, and project transactions. The frontend, built using HTML, CSS, and JavaScript, ensures an intuitive and responsive user experience. The backend processes client search queries, freelancer recommendations, and prebuilt project transactions, leveraging an AI-based matching system for optimal freelancer selection. The database schema, managed via Supabase, includes key entities such as Users (clients and freelancers), Freelancer Profiles (skills, experience, and availability), Projects (prebuilt solutions with descriptions and pricing). This architecture enables scalability and future feature expansions, such as enhanced AI recommendations and automated contract handling.

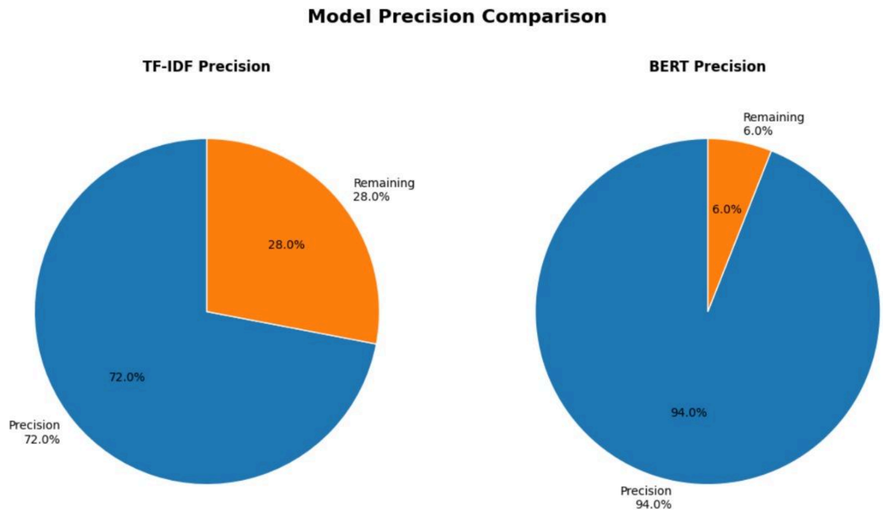

Figure 8.1: Model Precision

Figure 8.1 compares the precision of two models: TF-IDF and BERT. The TF-IDF model achieved a precision of 72%, whereas the BERT model significantly outperformed it with a precision of 94%. This indicates that BERT is more accurate in correctly identifying relevant results, minimizing false positives to a greater extent than the traditional TF-IDF approach. The difference highlights the advantage of using contextual embeddings in BERT over frequency-based representations in TF-IDF.

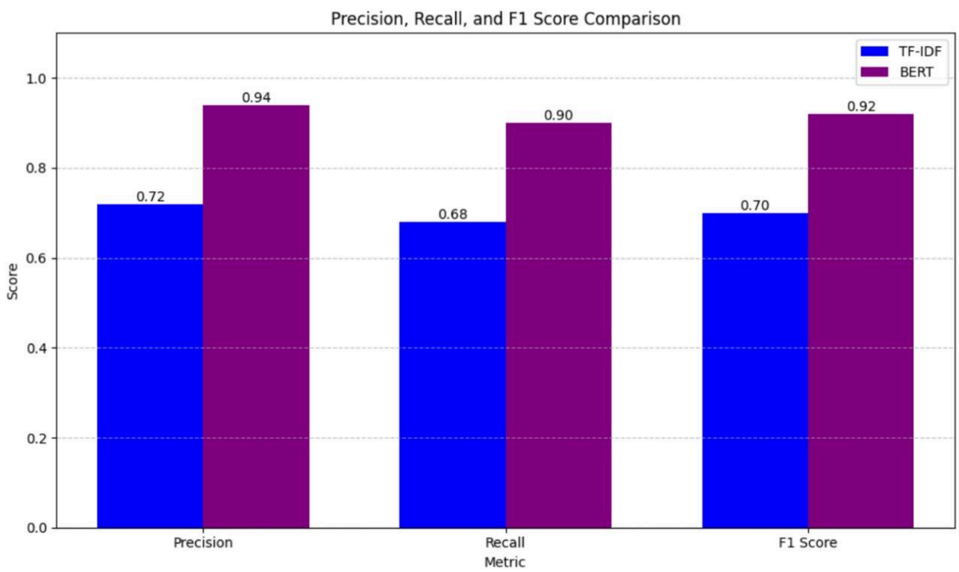

Figure 8.2: Precision, Recall & F1 Score

Figure 8.2 presents a bar chart comparing Precision, Recall, and F1 Score for both models. Across all three metrics, BERT consistently performs better:

Precision: BERT (0.94) vs. TF-IDF (0.72)

Recall: BERT (0.90) vs. TF-IDF (0.68)

F1 Score: BERT (0.92) vs. TF-IDF (0.70)

These results further validate BERT's superiority in both detecting relevant instances (precision) and retrieving all relevant ones (recall), culminating in a higher F1 score. This makes BERT a more balanced and effective choice for tasks requiring high classification performance.

Chapter 9

Conclusion

In conclusion, the MoodTracker application successfully addresses the growing need for real-time emotional support and personalized mental health management. By utilizing advanced sentiment analysis and AI technologies, the system offers users a comprehensive solution for tracking moods, receiving tailored recommendations, and engaging with an intuitive interface. The application not only simplifies emotional well-being management but also empowers users with insights and tools to foster long-term mental health improvements. This project demonstrates the potential of technology in transforming mental health care through innovative, user-centered design.

Chapter 10

Future Scope

The future development of Side-Kik can focus on enhancing user experience, improving AI-driven functionalities, and expanding platform capabilities. Key areas for future improvements include AI-driven contract management, blockchain-based transaction security, integration with third-party job marketplaces, and advanced analytics for freelancer performance insights. These enhancements will strengthen Side-Kik's position as a leading freelancing platform, providing a more efficient, secure, and intelligent experience for both clients and freelancers.

AI-Driven Contract Management: Implementing an AI-powered system for generating and managing contracts between freelancers and clients can streamline agreements, ensuring transparency and reducing disputes.

Blockchain-Based Transaction Security: Integrating blockchain technology for secure and transparent payments can enhance trust between clients and freelancers, ensuring secure transactions with immutable records.

Integration with Third-Party Job Marketplaces: Expanding Side-Kik's reach by integrating with platforms like LinkedIn or Upwork can allow freelancers to showcase their profiles across multiple job marketplaces, increasing visibility and job opportunities.

Advanced Analytics for Freelancer Performance: Developing analytics tools to track freelancer performance, client feedback, and project success rates can help freelancers improve their services and allow clients to make informed hiring decisions.

References

[1]Aditya Pawar et al., "A Freelancing Website (FrJobs)," International Journal of Research Publication and Reviews, Volume 4, Issue 10, Pages 56-63, 2023.

[2]Reshma Totare et al., "Freelancing Website Takenoko," International Research Journal of Modernization in Engineering, Technology and Science, Volume 5, Issue 5, Pages 78-85, 2023.

[3]Amit Narote et al., "Web App for Freelancing Developers and Designers," International Journal of Innovative Research in Technology, Volume 9, Issue 2, Pages 34-41, 2023.

[4]Bhupinder Kaur et al., "Online Freelancing Website," International Journal of Research in Engineering and Technology, Volume 9, Pages 12-18, 2021.

[5]Mr.Akshaya K M, Mr.Akarsha Kumar B H, Mr.Aravinda R, Ms.Veda N, “FREELANCING PLATFORM,” International Journal of Novel Research and Development, Volume 9, Issue 12 December 2024.